

POGRANICHNY, Russian Federation

WMO: 319150

Lat: 44.400N

Lon: 131.383E

Elev: 693

StdP: 14.33

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.2	-7.4	-21.1	1.8	-7.6	-18.1	2.1	-4.1	20.2	9.2	17.8	8.9	5.4	280	0.510

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.1	86.2	71.3	83.2	69.9	80.3	68.6	74.6	82.0	72.6	79.5	70.8	77.3	5.9	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
72.4	123.1	79.2	70.4	114.9	76.3	68.7	108.3	74.4	38.8	82.2	36.9	79.6	35.3	77.7	82.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
16.7	14.3	12.1		DB	-15.1	91.6	3.7	2.2	-17.8	93.2	-20.0	94.5	-22.1	95.7	-24.8	97.3
				WB	-15.5	77.3	3.5	2.4	-18.1	79.0	-20.1	80.5	-22.1	81.8	-24.7	83.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	41.2	7.0	14.5	27.7	43.5	55.3	63.7	70.0	69.3	59.2	45.2	25.9	10.7	(6)
		DBStd	23.40	7.51	8.53	9.16	7.47	6.37	5.66	4.78	4.70	5.98	7.73	10.96	8.15	
		HDD50	5393	1332	993	691	222	21	0	0	5	187	723	1219		
		HDD65	9091	1797	1413	1156	644	308	89	11	15	188	614	1172	1684	
		CDD50	2163	0	0	1	28	185	412	620	597	282	38	0	0	
		CDD65	386	0	0	0	0	6	51	166	147	16	0	0	0	
		CDH74	2831	0	0	0	17	176	488	1121	882	141	6	0	0	
		CDH80	679	0	0	0	1	45	120	316	181	16	0	0	0	
(14)	Wind	WSAvg	5.6	6.0	6.2	6.7	6.6	6.0	4.8	4.2	4.4	4.9	5.8	5.9	5.9	(14)
(15)	Precipitation	PrecAvg	24.3	0.4	0.4	0.6	1.4	2.3	3.5	4.2	5.2	2.9	1.8	1.0	0.6	(15)
(16)		PrecMax	34.5	1.4	1.5	2.0	4.3	4.4	8.5	10.5	12.3	5.9	6.4	3.0	2.0	(16)
(17)		PrecMin	12.4	0.0	0.0	0.0	0.4	0.7	0.3	1.3	0.7	0.1	0.1	0.1	0.0	(17)
(18)		PrecStd	4.9	0.4	0.4	0.5	0.9	1.1	2.0	2.2	2.6	1.6	1.3	0.7	0.5	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	31.0	43.2	59.1	76.5	86.3	87.7	90.6	87.9	82.2	73.4	59.2	38.8		(19)
(20)		0.4% MCWB	26.0	33.8	43.8	55.1	63.5	68.9	74.8	73.7	67.6	57.0	45.5	33.3		(20)
(21)		2% DB	26.3	37.1	52.1	69.3	78.8	83.5	87.0	84.5	77.8	67.7	52.7	32.6		(21)
(22)		2% MCWB	21.9	30.3	40.1	52.4	59.8	67.6	73.3	72.1	64.2	53.5	43.1	27.6		(22)
(23)		5% DB	22.4	32.3	46.5	63.7	73.6	79.6	83.6	81.7	74.6	63.0	46.7	27.7		(23)
(24)		5% MCWB	18.7	27.4	36.5	48.3	57.6	66.3	71.5	70.5	61.9	50.5	38.8	23.8		(24)
(25)		10% DB	18.5	28.4	41.8	58.6	69.3	75.8	80.1	79.1	71.5	58.8	41.6	23.6		(25)
(26)		10% MCWB	15.3	24.1	33.9	45.3	54.9	64.6	70.3	69.1	60.0	48.8	35.2	20.2		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	27.0	35.0	45.1	57.3	64.8	72.7	78.0	76.9	70.5	59.2	48.2	33.7		(27)
(28)		0.4% MCDB	30.7	41.1	58.1	72.6	81.9	83.5	87.2	85.0	77.3	69.1	56.3	38.5		(28)
(29)		2% WB	22.2	31.0	40.6	52.9	60.9	69.7	75.4	74.5	67.0	55.7	43.6	27.9		(29)
(30)		2% MCDB	26.1	36.3	50.5	67.5	76.4	79.4	83.2	81.1	73.8	64.0	51.8	31.9		(30)
(31)		5% WB	18.7	27.4	37.2	49.4	58.5	67.7	73.2	72.7	64.6	53.1	39.4	24.0		(31)
(32)		5% MCDB	22.3	32.3	45.3	62.0	71.7	77.0	80.3	78.7	71.2	60.0	46.2	27.6		(32)
(33)		10% WB	15.4	23.9	34.4	46.3	56.2	65.7	71.3	70.9	62.5	50.2	35.6	20.2		(33)
(34)		10% MCDB	18.4	28.5	41.2	57.0	67.0	74.2	78.3	76.5	68.9	57.1	41.3	23.6		(34)
(35)	Mean Daily Temperature Range	MDBR	17.1	19.3	19.2	21.9	21.2	19.1	16.1	16.6	21.9	22.1	18.6	16.6		(35)
(36)		5% DB MCDBR	20.7	23.5	26.0	31.2	30.6	25.9	22.4	20.6	25.4	28.7	25.0	21.2		(36)
(37)		5% WB MCWBR	17.2	18.3	16.5	18.1	15.0	12.8	10.3	9.7	12.7	16.2	17.4	17.5		(37)
(38)		5% WB MCDBR	20.6	23.3	24.8	29.8	28.6	22.9	18.6	17.3	20.9	24.8	24.6	21.1		(38)
(39)		5% WB MCWBR	17.3	18.5	16.7	18.1	14.8	12.5	10.0	9.4	11.8	15.2	17.9	17.6		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.323	0.372	0.458	0.566	0.585	0.566	0.528	0.485	0.438	0.456	0.386	0.337		(40)
(41)		taud	2.084	1.960	1.830	1.679	1.703	1.830	1.989	2.102	2.137	1.948	2.033	2.056		(41)
(42)		Ebn at Noon	240	247	239	224	226	231	239	245	246	215	211	218		(42)
(43)		Edh at Noon	32	44	58	73	74	65	55	47	42	44	34	30		(43)
(44)	All-Sky Solar Radiation	RadAvg	723	1077	1403	1457	1608	1702	1541	1475	1341	971	665	587		(44)
(45)		RadStd	52	51	113	125	157	184	156	130	108	69	53	31		(45)

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(46)
(47)	Regional (14 neighbors)	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(47)

Nomenclature: See separate page